

Conjugate heat transfer at an immersed interface: the scheme as implemented

As-built note — what the solver does today

Abstract

This note describes the conjugate-heat-transfer scheme *as it exists in the code*. It is deliberately narrower than the design note `conjugate_ibm.tex`, which surveys three source methods, derives the exact cut-face flux including its tangential term, works out a corner model, and weighs alternatives. None of that survey is repeated here. What is described here is what runs: one face coefficient, a fluid-fraction-weighted capacity, a time-step bound, a configuration surface with five hard guards, and two diagnostics. Where a term exists in the code but is switched off by default, it is stated as such together with the measurements that decided it (§9); where a piece of the design was never built, it is listed so that the reader is not misled (§10).

1 What is solved

The solid stops being a boundary condition and becomes a real unknown. Fluid and solid carry the same scalar variable, advanced by the same kernel, and differ only through two ratios that are 1 in the fluid by definition:

$$\kappa = \frac{k}{k_f}, \quad C = \frac{(\rho c)}{(\rho c)_f}.$$

In the solver's non-dimensional form, with $D_f = 1/(Re Pr)$ the fluid diffusivity, the transported scalar s obeys

$$C \partial_t s = -\nabla \cdot (\mathbf{u} s) + \nabla \cdot (\kappa D_f \nabla s) + C q, \quad (1)$$

with $\mathbf{u} \equiv \mathbf{0}$ and $\kappa = \kappa_s$, $C = C_s$ inside the body. At the interface the physical conditions are continuity of temperature and of normal flux,

$$s_f = s_s, \quad k_f \partial_n s_f = k_s \partial_n s_s, \quad (2)$$

optionally relaxed to a contact resistance R_c (§3). **Neither condition is imposed anywhere in the code.** They are consequences of a single face coefficient, which is the whole scheme.

2 Geometry: one stored field

The scheme reads exactly one geometric field, and it is one the solver already produced for the turbulence models:

$$\varphi = \pm \text{dwall}, \quad (3)$$

the distance to the surface, *signed* negative inside the solid. It is ghost-inclusive and pointwise exact from the geometry — no halo exchange is needed for it, because every value is computed from the surface, not from neighbours.

The sign comes from the cell-centred immersed-boundary marker: a cell centre lies in the solid exactly when $|\text{coef}(\text{VAR_P})| > 10^{20}$, because `set_ibm_coeff` writes `SOLID/Re` there and grades only fluid-centred cells. Both producers of `dwall` (the file-based path via the case file's `dwall_blocks`, and the analytic path) already supply what is needed; the conjugate mode adds **no new dataset and no case-file format change**.

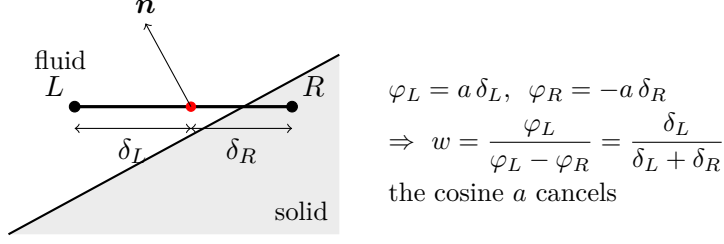


Figure 1: The obliquity lemma. φ stores the perpendicular distance, so the direction cosine a appears in both signed distances and cancels in their ratio: w is the arm fraction at any orientation, with no normal computed.

The obliquity lemma. This is why no normal vector is ever computed. Let the interface be locally a plane with unit normal $\mathbf{n}$, and consider a grid arm along $\mathbf{e}_d$ joining two cell centres at distance h_d , with $a = \mathbf{n} \cdot \mathbf{e}_d$ the direction cosine. Because φ stores the *perpendicular* distance,

$$\varphi_L = a \delta_L, \quad \varphi_R = -a \delta_R,$$

where δ_L, δ_R are the distances *along the arm* from each centre to the interface. The cosine therefore cancels in the ratio:

$$w = \frac{\varphi_L}{\varphi_L - \varphi_R} = \frac{\delta_L}{\delta_L + \delta_R} \quad (4)$$

is the true arm fraction *at any orientation*. That is the entire reason the scheme needs no new data.

3 The face coefficient — the whole scheme

At a face whose two cell centres straddle the interface ($\varphi_L \varphi_R < 0$), the face diffusivity becomes the distance-weighted harmonic mean on w :

$$k_{\text{face}} = \frac{D_f}{\frac{w}{\kappa_L} + \frac{1-w}{\kappa_R} + R_c \frac{D_f}{h_d}} \quad (5)$$

Every other face keeps today's kernel line verbatim, $k_{\text{face}} = \kappa_L D_f$. That is the complete interface treatment: one sign test, one weight, one reciprocal, evaluated inside the flux line the kernel already had.

What it inherits:

- **Exact in one dimension.** It is the series resistance, so it is exact for a grid-aligned interface at *any* cut position, and exact at any orientation wherever the local gradient is interface-normal.
- **Conservative.** It is a face flux — the neighbour subtracts the same number — so $\sum C s \Delta V$ is conserved to round-off.
- **Correct in both limits.** $\kappa_R \rightarrow \infty$ recovers the isothermal wall, $\kappa_R \rightarrow 0$ zero flux.
- **Inert when unused.** An uncut face takes the else-branch, which is the pre-existing arithmetic operand for operand, so a non-conjugate run is bit-identical.

Grazing guard. w is unreliable for arms nearly *tangent* to the surface, where $\varphi_L - \varphi_R \rightarrow 0$. The implementation therefore tests the direction cosine, which (4) supplies for free as $a = |\varphi_L - \varphi_R|/h_d$, and falls back to $w = \frac{1}{2}$ below a threshold. Such arms carry little of the interface flux, so the fallback costs nothing measurable.

Contact resistance. R_c enters (5) as a third resistance in series, which is what it physically is. It is exact: a gate with $R_c = 0.25$ reproduces the analytic two-material solution to 0.0.

4 The discrete scheme

4.1 Face loop

The six face diffusivities of a pressure cell are built with (5) and the flux divergence is formed as usual. Nothing else in the diffusion stencil changes.

4.2 Convection

Convection is hard-masked on solid *and* cut faces, and — this is the part that is easy to get wrong — the skew-symmetric term’s divergence is built from the *same* masked velocities. That is what keeps a uniform scalar exactly preserved, in the solid as well as the fluid.

4.3 Capacity at cut cells

The flux divergence is divided by the cell’s own volumetric capacity. A cut cell physically contains both materials, so the capacity is **fluid-fraction weighted**:

$$C_{\text{cell}} = f + (1 - f) C_s, \quad \text{rhs} = \frac{-\text{conv} + \text{diff} + f q_f + (1 - f) C_s q_s}{C_{\text{cell}}}, \quad (6)$$

with f the fluid volume fraction of the cell. The volumetric source is weighted with the same fraction, as the power density it is.

f is pure geometry, built once at init beside φ from the closed plane-in-box form and stored as `vfrac`; it rides the snapshot so a diagnostic can state the invariant the solver actually conserves. The common approximation $\text{clip}(\frac{1}{2} + \varphi_c/h)$ is not an approximation of that form but its degenerate limit — the one a grid-aligned wall lands in.

Why the weighting is not cosmetic. It cannot change any steady state, because it divides a residual that vanishes there. It is what makes the *transient* second order: with a pointwise capacity a cut cell carries one material’s entire heat capacity, an $O(1)$ error on an $O(h)$ band, i.e. first order. Measured on a two-material eigenmode, against the pre-C3 binary on the same inputs:

	field-error order	decay-rate order
fluid-fraction weighted (6)	1.99, 2.00	1.99, 2.00
pointwise capacity	1.71, 1.09	1.28, 0.54

4.4 What is deliberately absent

There is **no penalization** in the conjugate mode. Its solid cells are unknowns, not boundary values, so the volume-penalization term that the `dirichlet` mode applies is simply not entered. Likewise ν_t is admitted neither in the solid nor at a cut face.

5 Time step

Two findings, both measured, are built into the explicit bound.

The limit is not the maximum material diffusivity. A cut face’s k_{face} can approach $\max(\kappa_L, \kappa_R)$ while feeding the cell on the *other* side, whose capacity is the other material’s. A fluid cell against a $\kappa_s = 1000$ solid therefore sees $1000\times$ the fluid rate *even when* $\alpha_s = \alpha_f$. The rate is consequently assembled from the *actual face coefficients* rather than from $\max(\kappa/C)$ — which is also some $500\times$ less conservative at $w = \frac{1}{2}$.

A cut cell attains the Gershgorin bound. The uniform interior never excites $\rho \leq 2A_{ii}/C_i$; a cut cell does. Its rate is therefore raised: the implementation uses

$$\text{share} = \begin{cases} 2 & \text{at a cut cell,} \\ 6 & \text{otherwise,} \end{cases}$$

i.e. $\text{dt} \times \frac{2}{3}$ *at cut cells only*. The history is worth keeping: the first version used 6 everywhere, C1 halved it to 3, and C2 then measured the default sitting at 96 % of the bound and found an oblique high-contrast case that went to NaN there. At $\text{share} = 2$ the margin is $1.56\times$ and that case runs.

6 Order of accuracy

The scheme is **second order where the interface is flat or the local temperature gradient is interface-normal, and first order at an obliquely cut or curved interface carrying a genuine tangential gradient**. The condition is geometric, not a matter of resolution. It is exact — not merely second order — when the interface normal is a grid direction, which covers every flat-wall case including all of the channel validation in §11. At an obliquely cut or curved interface carrying a tangential gradient the cut-face *flux* error does not converge at all; the conservative cell balance damps that into a *first-order* solution error — measured below on both geometries, so “does not converge” is a statement about the flux and never about the scheme. Each statement is measured.

regime	what is measured	result
Interface normal along a grid direction, <i>any</i> cut position within the cell; steady	1D two-material slab, cut swept through a full cell, κ_s over five decades	exact , $\leq 9.7 \times 10^{-15}$
The same, <i>transient</i>	decaying two-material eigenmode; field error and decay rate	1.99, 2.00
The same, but with a <i>pointwise</i> cut-cell capacity	the same eigenmode, archived pre-C3 binary	1.71, 1.09 — first order
Oblique face, gradient purely interface-normal ($r = 0$)	cut-face normal flux against the analytic conjugate field	exact , $\sim 10^{-10}$ (geometry floor)
Oblique face, tangential ratio $r > 0$ — the cut-face <i>flux</i>	the same, at $h = 1/64, 1/128, 1/256$	error $\propto r$, independent of h
The same, the <i>solution</i>	oblique-plane conjugate BVP, solved and refined	first order (1.00 in the interface band)
Curved interface, the <i>solution</i>	cylinder in a harmonic far field, $n = 64 \dots 512$	first order (0.999)

The two ways to be second order. The series resistance (5) is the *exact* solution of the one-dimensional conjugate problem, so wherever the problem is locally one-dimensional the interface contributes no error at all and the scheme is simply the underlying second-order discretisation. Two distinct situations qualify, and the first is not an approximation:

- the interface normal is a grid direction — every flat-wall case, including all of the channel validation in §11. The cut position is irrelevant: the slab gate sweeps it through a whole cell and the error stays at round-off;
- the local gradient is interface-normal, $r = |\nabla_t T|/|\partial_n T| \rightarrow 0$, which is the resolved-DNS premise the scheme was designed on.

Where it is not second order, and it is worse than first. At an obliquely cut face the baseline’s implied normal flux is $q_n + h_d s_t/(a\mathcal{R})$, and since $\mathcal{R} \sim h_d/k$ the error is $\sim k s_t/a$ — proportional to a *physical* gradient and carrying no power of h . Measured at $\theta = 30^\circ$, $\kappa_s = 10$, the rms flux error is 3.56×10^{-2} , 3.77×10^{-2} , 3.59×10^{-2} at $h = 1/64, 1/128, 1/256$: an observed order of -0.01 across a fourfold refinement, i.e. flat. It scales exactly linearly with r (3.56×10^{-2} , 3.56×10^{-1} , 3.56 at $r = 0.01, 0.1, 1$).

So at such a face the error **plateaus at** $O(r)$ rather than reducing. This is a consistency error on a codimension-one set, and how much of it survives into the solution norm has now been measured.

The solution order at an oblique plane: one. On the manufactured oblique plane of §9 ($\theta = 30^\circ$, $\kappa_s = 10$, $r = 0.01$), solved and refined over $n = 64, 128, 256$, the interface-band L_2 error falls at order **1.00** (2.10×10^{-6} , 7.23×10^{-7} , 2.52×10^{-7} ; globally 1.53, since the band is a shrinking fraction of the domain). So the $O(r)$ flux plateau costs one order, not convergence. For contrast, the multiplier of §9 applied with the same discrete s_t is $25\times$ *worse* than the baseline here and still first order, while a converging s_t makes the solution exact to round-off (1.5×10^{-13}): on a plane the series resistance is already exact, so once the tangential term is right the interface contributes nothing at all.

The solution order on a curved interface: one. *Measured 2026-09-15, and it is a correction to what this section previously left open.* The discrete conjugate BVP is solved on a conducting cylinder in a harmonic far field — exact solution known in both materials, Dirichlet data from it, no source — and refined. Every scheme here is linear in T with purely geometric coefficients, so its matrix is recovered exactly by graph colouring (checked against the operator to 2×10^{-16}) and solved directly: no iterative tolerance enters the error.

κ_s	as implemented (§??)		with a converging s_t	
	L_2	L_∞	L_2	L_∞
0.01	1.11	0.94	2.00	1.60
0.1	1.10	0.92	2.04	1.62
2	1.07	0.86	1.87	1.60
10	1.00	0.91	1.92	1.78
100	1.36	1.34	1.97	1.47
1000	1.89	1.65	1.99	1.49

Grids $n = 32, 64, 128, 256$ (the $\kappa_s = 10$ row over $n = 64 \dots 512$, an eightfold range, where the as-implemented order is 0.999). A second geometry — the dipole rather than the quadrupole — reproduces it: 1.03–1.09 against 2.09–2.11.

Three things follow. First, the conservative finite-volume structure *does* damp the non-converging face error, by exactly one power of h : an $O(1)$ flux error on a codimension-one set becomes a first-order solution error, not a stalled one. Second, **first order is the honest description of the scheme at a curved conjugate interface** over the whole physically interesting contrast range. Third, the baseline recovers second order only as $\kappa_s \rightarrow \infty$, and for a physical reason rather than a numerical one: $|\nabla_t T|$ at the interface scales as $2/(1 + \kappa_s)$, so in the isothermal limit the term being dropped vanishes on its own. The scheme’s *worst* case is therefore moderate contrast, not high — the opposite of what the face-flux measurements suggest.

The right-hand columns are not implemented; they are what a converging tangential estimate delivers, measured on the same problems (`validation/conjugate/README.md`, stages 0–1b).

Why that was accepted. Because $r \sim 10^{-2}$ in a resolved thermal layer, the plateau sits near 1%, and because the measured crossover is $r^* \approx 0.027$: below it the corrected flux is *worse* than the baseline (§9). Restoring formal second order at oblique faces is therefore not a matter of switching the existing term on.

The capacity is what secures the transient order. Note the third row of the table: the spatial treatment above is unchanged there, and the order still falls to first. A cut cell carrying one material’s whole heat capacity is an $O(1)$ error on an $O(h)$ band, and the decay rate μ is a global functional of the capacity distribution, so it is visible in every norm.

7 Configuration surface

key	meaning
[scalar.N] <code>ibm_wall = conjugate</code>	selects the mode (beside <code>dirichlet</code> , <code>adiabatic</code>)
<code>solid_k</code>	κ_s
<code>solid_rhocp</code>	C_s
<code>solid_init</code>	initial solid value
<code>solid_source</code>	volumetric source inside the solid
<code>contact_resistance</code>	R_c in (5)
<code>tangential_correction</code>	the C2 term, default off (§9)
[scalar] <code>indicator_interval</code>	the free error indicator
[scalar] <code>heat_interval</code>	the interface-heat diagnostic

`dirichlet` and `adiabatic` were left untouched: re-expressing them through the new arithmetic could not have been bit-exact, and bit-exactness of the existing modes was worth more than the shared code path.

Five hard configuration errors. Each of these silently produces a wrong answer if allowed, so each stops the run with a message naming the fix: `remove_solid = true` (the solid must exist); `refine_body` without `keep_buried` (same); `ibm_value` on a conjugate scalar (there is no imposed value); a `solid_*` key on a non-conjugate scalar; no immersed body at all; and wall functions. A sixth check is geometric rather than configurational: the 2:1 precondition — a cut face may not sit on a coarse/fine block face, because the coefficient is a same-level arm — is *verified* at init, not assumed.

8 Diagnostics

Interface heat (the Nusselt number). The conjugate branch sums the scheme’s *own* face flux over the cut faces, with the penalization column explicitly zero — there is none in this mode, and `coef_p` is finite in a graded fluid cell, so writing the zero explicitly forestalls a double count. Gated three ways: against a closed form (4.6×10^{-13}), against an independent Python transcription (3.6×10^{-16}), and against a control-volume budget with the flow on that falls like dt^2 and does not move with `niter`.

The error indicator. `indicator_interval` reduces $e_{\text{face}} = h_d s_t / (T_R - T_L)$ over the cut faces (max, rms, count). This is the closed form of the relative error the scheme makes by dropping the tangential term, so it is a genuine *a-posteriori* measurement rather than an assumption, and it is free. It ships enabled.

9 Present but disabled: the tangential correction

The exact cut-face flux carries a second term, $s_t (k_{\text{loc}} - k_{\text{face}})$, which the scheme above drops. It *is* implemented, as its own flux divergence added after the baseline term — so it is inert by construction when off, not by a `+0.0` argument — and it is **default off**. Three independent measurements decided that:

1. a crossover ratio $r^* \approx 0.027$, which the DNS-like $|\nabla_t T|/|\partial_n T| \sim 10^{-2}$ sits below;
2. an $r = \infty$ boundary-value problem in which the corrected *face flux* is exact to 10^{-11} while the corrected *field* is $2.6\times$ *worse* — a finite-difference divergence wants the face *average*, not the midpoint value;
3. decisively, a cylinder at $\kappa_s = 10^3$ where the correction is $40\text{--}65\times$ worse than dropping the term, because $|\nabla_t T| \propto 2/(1 + \kappa_s) \rightarrow 0$ in the isothermal limit while the discrete estimate of it does not.

The measured fix for (3) — k_{area} , the face area-weighted mean, which removes the paradox exactly on a plane — is recorded but deliberately not implemented: it cannot rescue the feature, because the binding constraint is the s_t estimate on curved interfaces. **Anyone reopening this starts from s_t , not from the multiplier.**

10 Not implemented

For the avoidance of doubt, the following appear in the design note and do *not* exist in the code: the corner/wedge (COCO) model for conducting sharp corners; the escalation back to the full exact flux as a supported mode; and any pluggable local-solution interface. The corner model rests on the same pointwise-flux premise that measurement (2) above falsified, so it should not be started without first settling the s_t estimate.

11 Validation status

Manufactured solutions and identities. A 1D two-material slab with the cut swept through a full cell and κ_s over five decades: $\max |\theta - \theta_{\text{exact}}| \leq 9.7 \times 10^{-15}$ over 28 pairs, and w itself 2.2×10^{-14} against the analytic cut, rebuilt from the case file without the solver. Capacity-independent steady state; contact resistance exact; $\sum C\theta \Delta V$ drift 1.2×10^{-16} relative with the flow on. 1 rank = 4 ranks and CPU = GPU at `max_abs` 0 on both geometry paths.

Every `ibm_wall` $\neq$ `conjugate` run and every `[scalar] count = 0` run bit-exact against the pre-feature binaries.

Turbulence. A conjugate channel at $Re_\tau = 149$, $Pr = 0.71$, matched to Flageul et al. (2015) in thermal problem, Reynolds number, grid resolution ($\Delta y^+ = 0.49 \rightarrow 4.87$, $\Delta x^+ = 8.4$, $\Delta z^+ = 4.2$), solid thickness ($d^+ = 149$) and outer-wall boundary condition (imposed flux), compared against *their published raw data* over a clean 29 684 wall-unit window. Agreement, as a percentage of each quantity’s own peak:

U^+	0.4 %	$-\overline{u'v'}$	0.4 %
θ^+	1.2 %	$\overline{u'T'}$	1.3 %
$\overline{u'^2}$	0.9 %	$\overline{v'^2}$	1.8 %
$\overline{w'^2}$	3.8 %	$\overline{T'^2}$	2.4 %

The near-wall variance peak is 5.98 against their 5.94, +0.7%, and θ_τ is κ_s -independent across the sweep to 0.003 %. The velocity comparison matters independently: it is the first against a trustworthy reference, so the thermal agreement is not resting on a compensating error in the flow.

Inside the solid. The scheme is also checked where the solid is the answer rather than a boundary condition — the variance profile *through* the slab, against their `fluctb` data:

depth y^+	−77	−102	−145 (outer face)
ours	0.0113	0.0075	0.0063
theirs	0.0113	0.0083	0.0073

This one carries a methodological warning worth more than the numbers. The first attempt read $8\times$ their value at the outer face, and the cause was neither the scheme nor the grid: the solid diffusion time is $d^2/\alpha_s \approx 106$ time units and the averaging window was shorter than *one*, so the statistics were dominated by the initial transient. Splitting the variance into a within-plane (spatial) and a plane-mean (temporal) part isolates it — the temporal term falls by a factor 45 when the transient is dropped. **Start conjugate statistics only after one solid diffusion time, and seed the solid rather than waiting for it:** with an imposed flux at both outer faces the solid’s mean level is a pure-Neumann null-space mode, so it can be set in closed form instead of relaxed.